

DEVOPS THEORY NOTES (EXAM READY)

◆ What is DevOps?

DevOps is a combination of **Development and Operations**. It is a set of practices that improves collaboration between developers and IT operations teams to deliver software **faster, more efficiently, and with fewer errors**.

☞ It focuses on **automation, continuous integration, and continuous delivery (CI/CD)**.

◆ Why Do We Need DevOps?

DevOps is needed to solve problems in traditional software development:

- Development and operations teams work separately
- Software delivery is slow
- Bugs are detected late

☞ DevOps improves:

- Speed of delivery
 - Collaboration
 - Software quality
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◆ Benefits of DevOps

- Faster software releases
 - Better collaboration between teams
 - Reduced errors and bugs
 - Continuous feedback and improvement
 - More reliable systems
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◆ What Does a DevOps Engineer Do?

A DevOps Engineer:

- Automates software development processes
 - Manages CI/CD pipelines
 - Works with tools like Git and Docker
 - Monitors system performance
 - Ensures smooth deployment of applications
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◆ DevOps Workflow

DevOps workflow shows how software moves from development to production:

☞ Steps:

- Code
 - Build
 - Test
 - Release
 - Deploy
 - Monitor
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◆ DevOps Lifecycle

The DevOps lifecycle is a continuous process:

☞ Plan → Code → Build → Test → Release → Deploy → Monitor

It ensures **continuous improvement and faster delivery**.

◆ DevOps Pipeline

A DevOps pipeline is an automated sequence of steps used to:

- Build code
- Test code
- Deploy applications

☞ It reduces manual work and errors.

◆ CI/CD Introduction

CI/CD stands for:

- **Continuous Integration (CI)** → code is regularly merged and tested
- **Continuous Delivery/Deployment (CD)** → code is automatically deployed

☞ It helps deliver updates quickly and reliably.

◆ What is Version Control?

Version control is a system that tracks changes in code over time.

☞ It allows:

- Multiple developers to work together
 - Code history tracking
 - Easy rollback to previous versions
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◆ What is Git?

Git is a **distributed version control system** used to manage code.

☞ It allows:

- Tracking changes
 - Branching and merging
 - Collaboration among developers
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◆ The Standard Git Flow

Git flow is a structured way of working with branches:

🔗 main → dev → feature branches

- Developers create feature branches
 - Merge into dev
 - Then merge into main
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◆ What is a Branch?

A branch is a **separate version of code**.

🔗 It allows developers to work independently without affecting the main code.

◆ Why Branching is Essential

- Supports teamwork
 - Prevents breaking main code
 - Allows testing new features safely
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◆ Branch Lifecycle

1. Create branch
 2. Work on feature
 3. Commit changes
 4. Merge into main branch
 5. Delete branch
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◆ Branch Commands

```
git status
git checkout -b feature
git add .
git commit -m "msg"
git push origin feature
```

◆ What is a Merge Conflict?

A merge conflict occurs when two branches modify the same part of a file.

☞ Git cannot decide which change to keep.

◆ Pull Requests

A Pull Request (PR) is used to:

- Review code
- Discuss changes
- Approve before merging

☞ It acts as a **quality control step**.

◆ Pull Request Workflow

1. Create branch
 2. Push code
 3. Open PR
 4. Code review
 5. Fix issues
 6. Approve
 7. Merge
 8. Delete branch
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◆ What are Permissions and Why Do They Exist?

Permissions control access to files and directories.

☞ Types:

- Read (r)
- Write (w)
- Execute (x)

☞ Purpose:

- Security
 - Prevent unauthorized access
 - Control file usage
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◆ What is Docker?

Docker is a **containerization platform** that allows applications to run in isolated environments called containers.

◆ What Problem Docker is Resolving?

Problem:

☞ “Works on my machine” issue

Different systems have different environments.

Solution:

☞ Docker provides **same environment everywhere**

◆ What is a Dockerfile?

A Dockerfile is a script that contains instructions to build a Docker image.

☞ Example tasks:

- Set base image
- Copy files
- Install dependencies
- Run application

System Monitoring Commands

- **CPU Usage** → `top -bn1 | grep "Cpu"`
- **RAM Usage** → `free -h`
- **Disk Usage** → `df -h`

◆ User & Directory Commands

- **Check Username** → `whoami`
- **Current Directory** → `pwd`
- **Change Directory** → `cd folder_name`

◆ File & Folder Commands

- **List Files** → `ls`
- **Create Folder** → `mkdir folder_name`
- **Create File** → `touch file.txt`
- **Delete File** → `rm file.txt`

◆ File Viewing & Searching

- **View File Content** → `cat file.txt`
- **Search in File** → `grep "word" file.txt`

◆ Permissions

4 = r

2 = w

1 = x

- **Change Permissions** → `chmod 755 file.sh`

☞ Meaning:

- Owner → rwx
- Group → r-x
- Others → r-x

◆ Script / Command Result

- **Check Last Command Status** → `echo $?`

☞ 0 = success

☞ non-zero = error